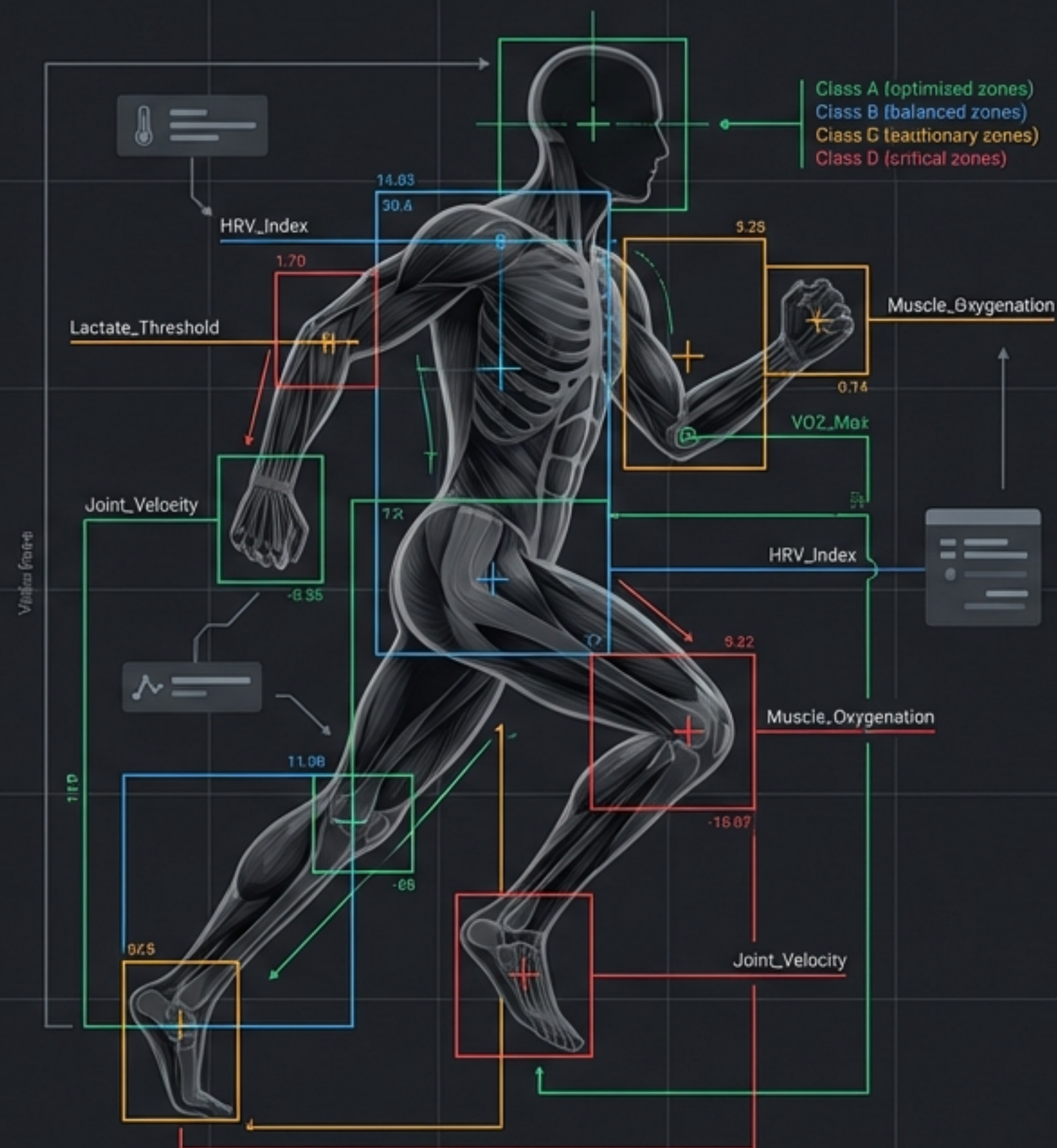


Predicting Human Performance

Decoding the physiological signals of fitness using machine learning.



Based on National Fitness Program Analytics (13,393 records, 12 physiological markers).

Executive Summary: The Signals and the Noise



1. The Biological Bottleneck

The human-designed 4-class grading system is fundamentally flawed. Classes B and C overlap so heavily they are virtually indistinguishable physically.



2. Algorithmic Mastery

XGBoost cuts through the physiological noise, emerging as the undisputed champion model.

89.0%
Accuracy
(Binary Classification)

77.5%
Accuracy
(4-Class System)



3. The Signal Engine

Advanced feature engineering and SHAP analysis reveal that relative core strength and `gripPerLeanMass` and body composition dictate ultimate physical performance far more than raw age.

Data Triage: Separating Biology from Error

Raw Ingestion: 13,393 Rows

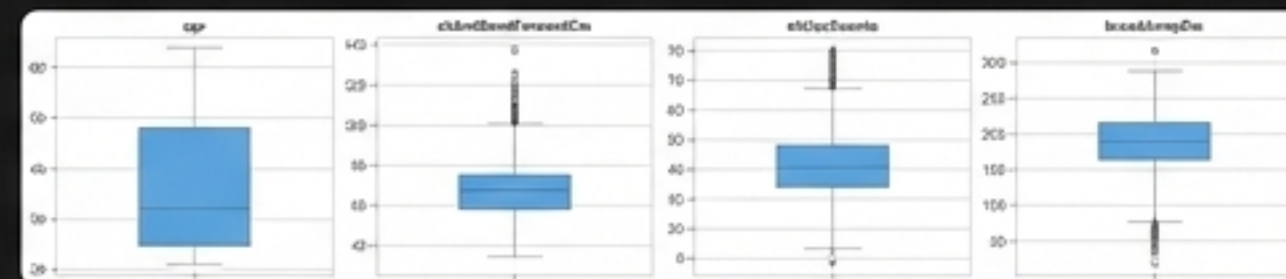
⚠ Filter 1: Medical Impossibilities (Removed)

- **Clinically Dead:** BP 0/0 mmHg, Diastolic < 20
- **Thermodynamic Anomalies:** Diastolic $\geq$ Systolic
- **Incompatible with Life:** BMI < 14, Body Fat < 5% or > 50%

~120 rows dropped (0.9%)

✅ Filter 2: Valid Biological Variance (Retained)

Extreme strength or flexibility represents natural human variance, not measurement error. All Interquartile Range (IQR) outliers were purposefully retained.



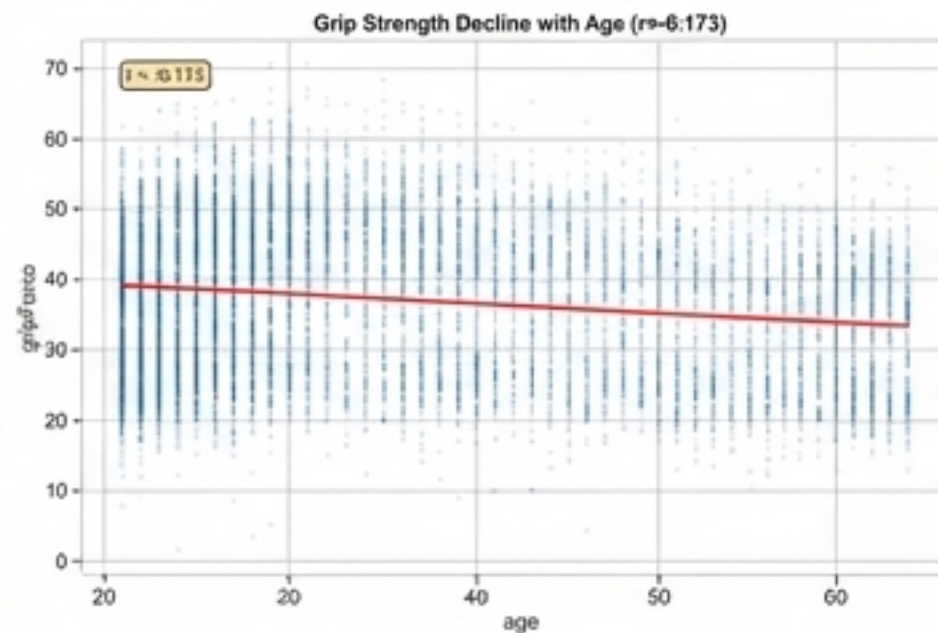
Clean Pipeline Output:
~13,270 Rows

Insight: A clean dataset is defined not by removing statistical extremes, but by eliminating physiological impossibilities.

The Human Blueprint: Physiological Ground Rules

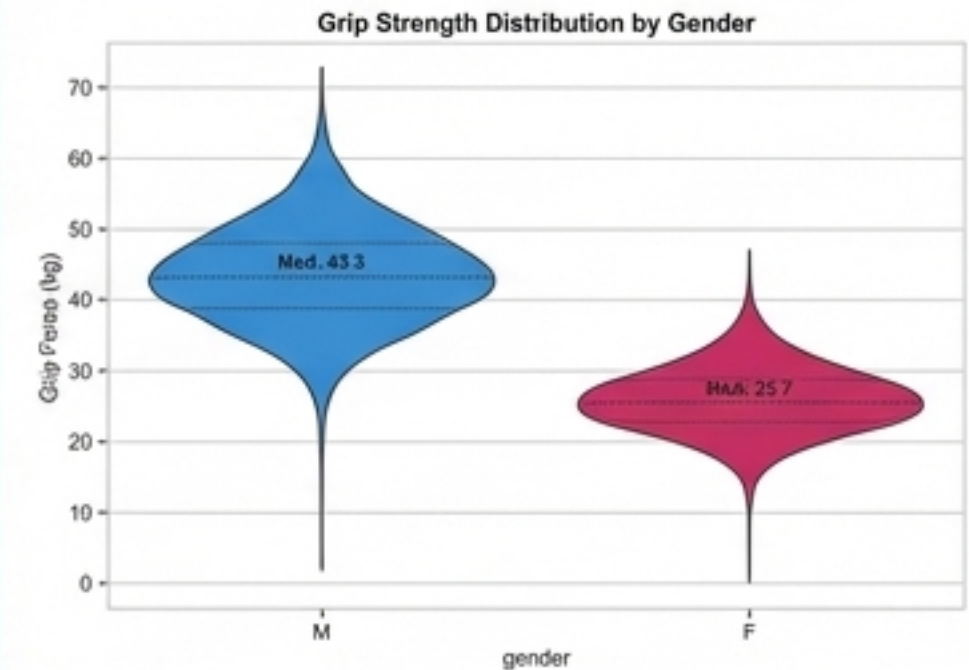
The Age Tax

All fitness metrics degrade with time, accelerating sharply after age 50. Core endurance (sitUpsCounts) suffers the steepest penalty.



The Gender Divergence

Males outperform females in raw grip strength by ~15 kg, while females carry 8-10% more essential body fat. Flexibility is universally comparable.

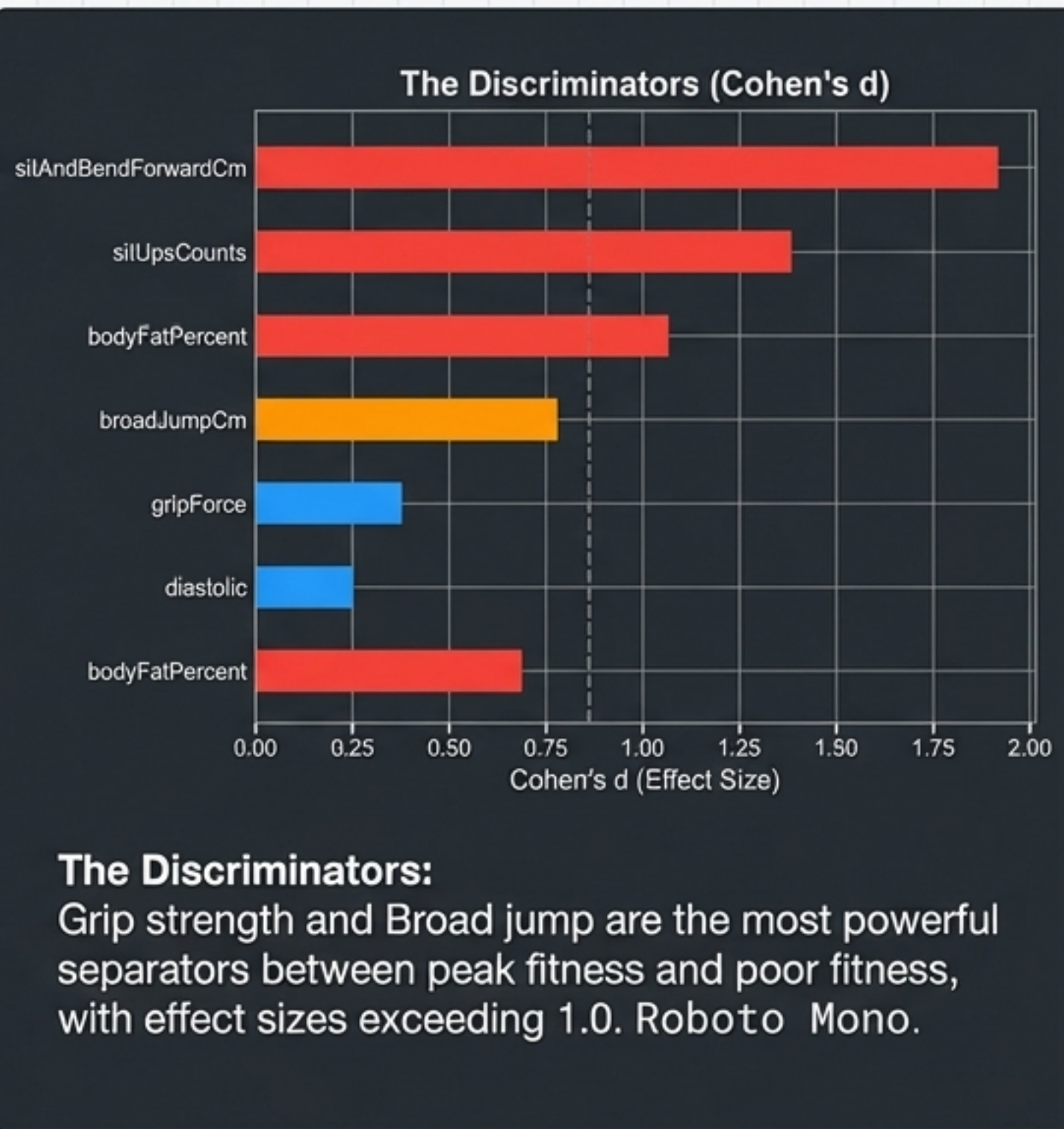
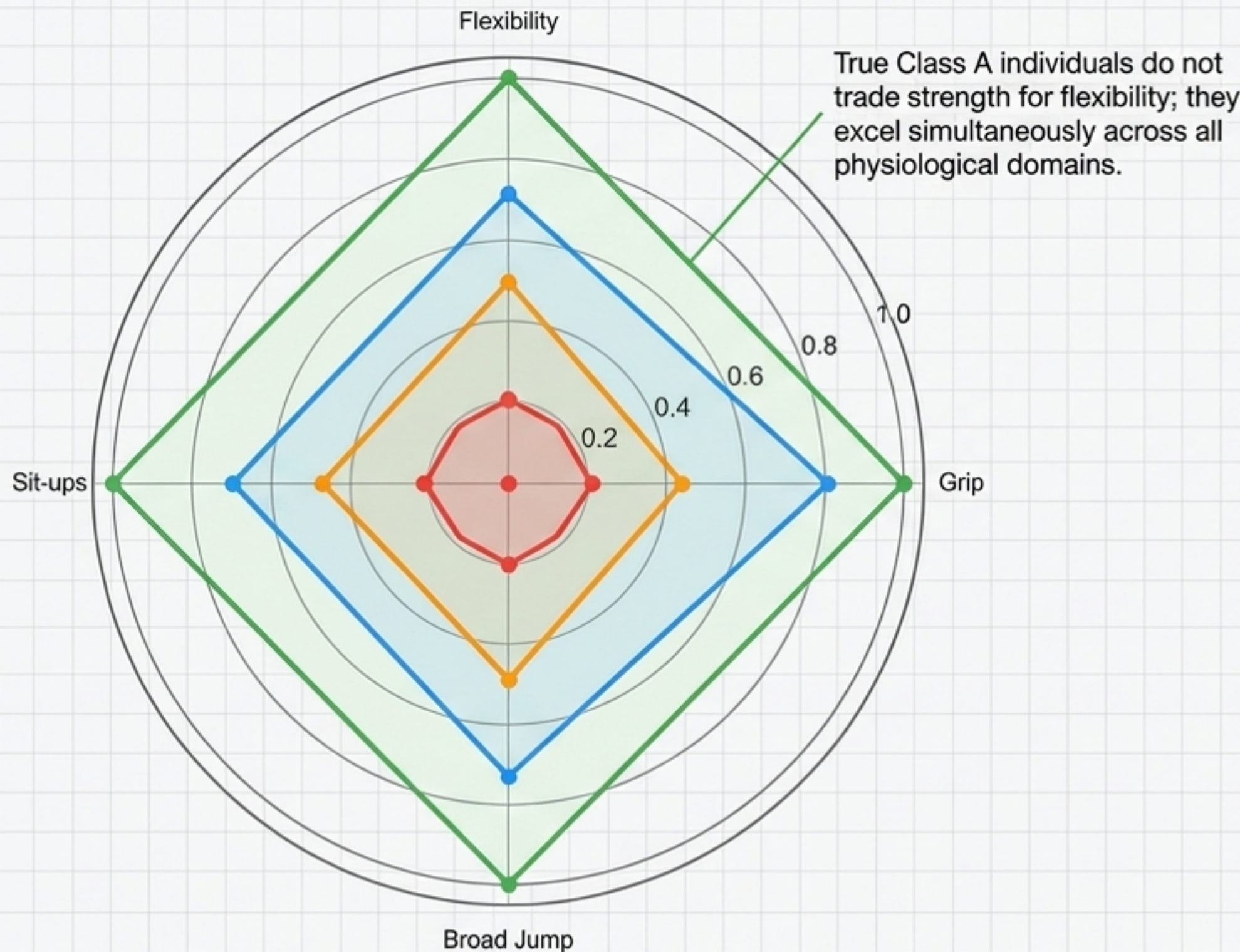


The Mass-Power Link

Weight strongly correlates with raw localized force. Heavier individuals generate significantly higher absolute grip force ($r = +0.43$).

Insight: Human fitness follows strict, predictable physiological rules across age, gender, and mass prior to any algorithmic intervention.

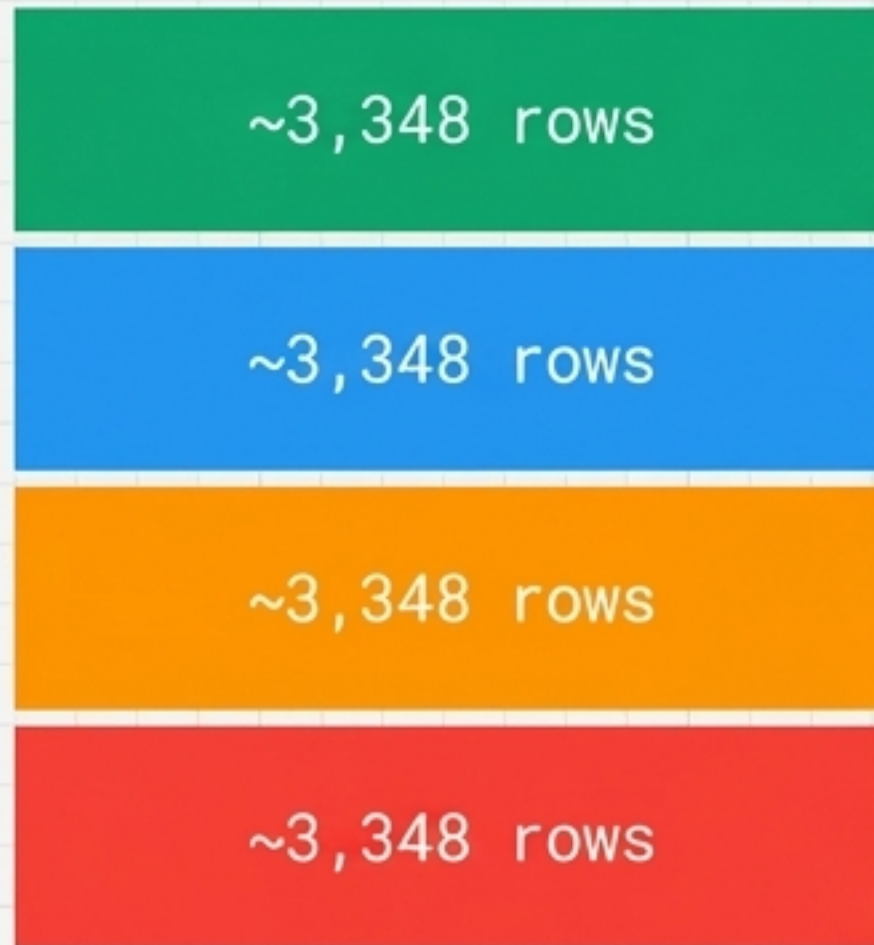
The Anatomy of a Class A Performer



Insight: Elite performance is fundamentally holistic. The gap between the best and the worst is widest in explosive power and grip strength.

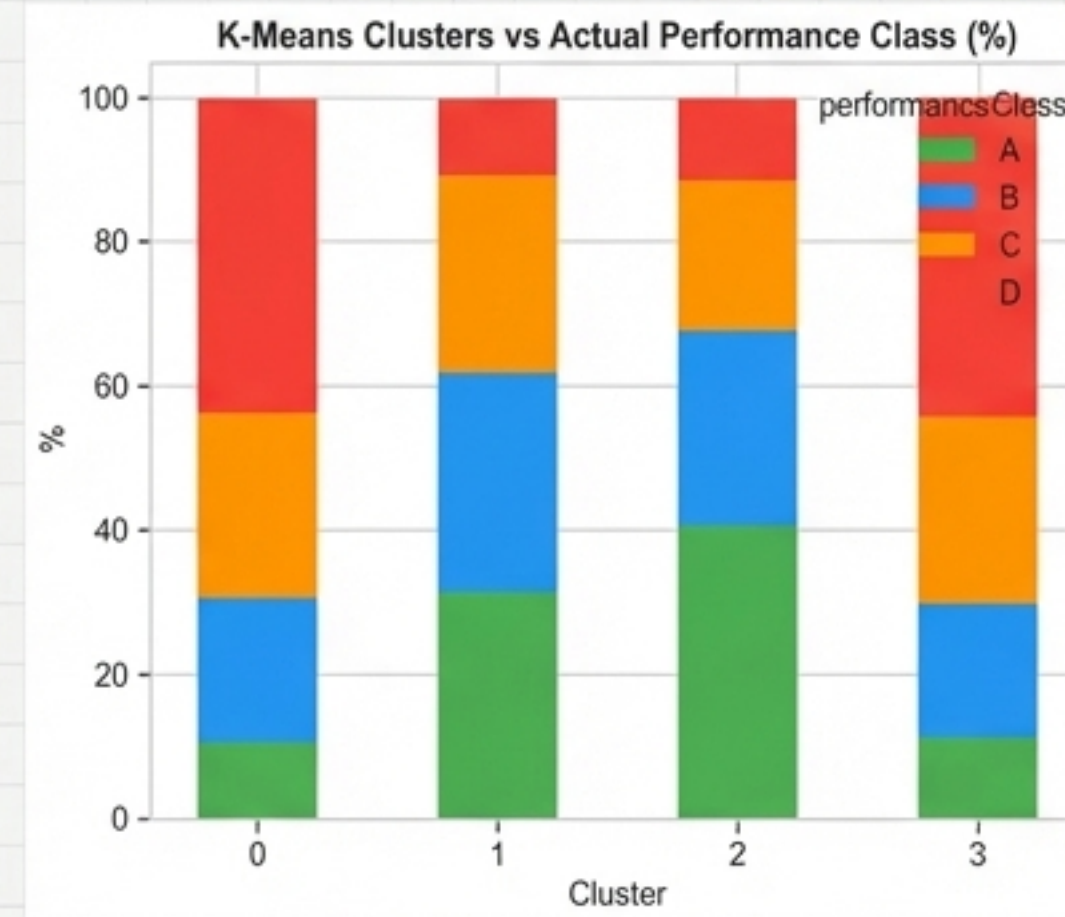
The Central Conflict: Artificial Design vs. Biological Reality

The Assessment Flaw (Artificial Quotas)



The national testing handbook forces a strict quartile-based mathematical divide where a biological one doesn't exist, assigning exactly 25% of individuals to each bucket.

The Collision of Average (Biological Reality)



Unsupervised learning struggles to separate Classes B and C. Physiologically, an above-average Class C and a below-average Class B are statistically indistinguishable.

Insight: The primary bottleneck for any predictive model is not the measurement data, but the fundamentally flawed, human-drawn boundary between the 'average' Classes B and C.

Target Architecture Matrix: Engineering the Objective

4-Class (The Baseline)

Classes involved: A, B, C, D

Rationale: Original labels required by the assessment handbook.

77.5%

3-Class (The Biological Correction)

Classes involved: High (A), Average (B+C), Low (D)

Rationale: Merges the statistically indistinguishable middle classes.

83.4%

2-Class (The Binary Standard)

Classes involved: Good (A+B), Poor (C+D)

Rationale: Practical pass/fail threshold for physical standards.

89.0%

Insight: Restructuring the target to align with physiological reality rather than artificial quotas unlocks a massive +11.5% leap in predictive power.

Signal Extraction: Feature Engineering

bmi

$$\text{weight} / (\text{height}/100)^2$$

Standard metabolic indicator combining mass and height into a single dense metric.

sitUpsCounts_zGender

$$\text{gender_normalized_z_score}$$

Strips away the baseline biological differences between sexes to measure pure, relative core endurance.

gripPerLeanMass

$$\text{gripForce} / (\text{weight} \times (1 - \text{fatPct}/100))$$

Calculates 'pound-for-pound' strength. A vastly superior predictor to raw absolute grip force.

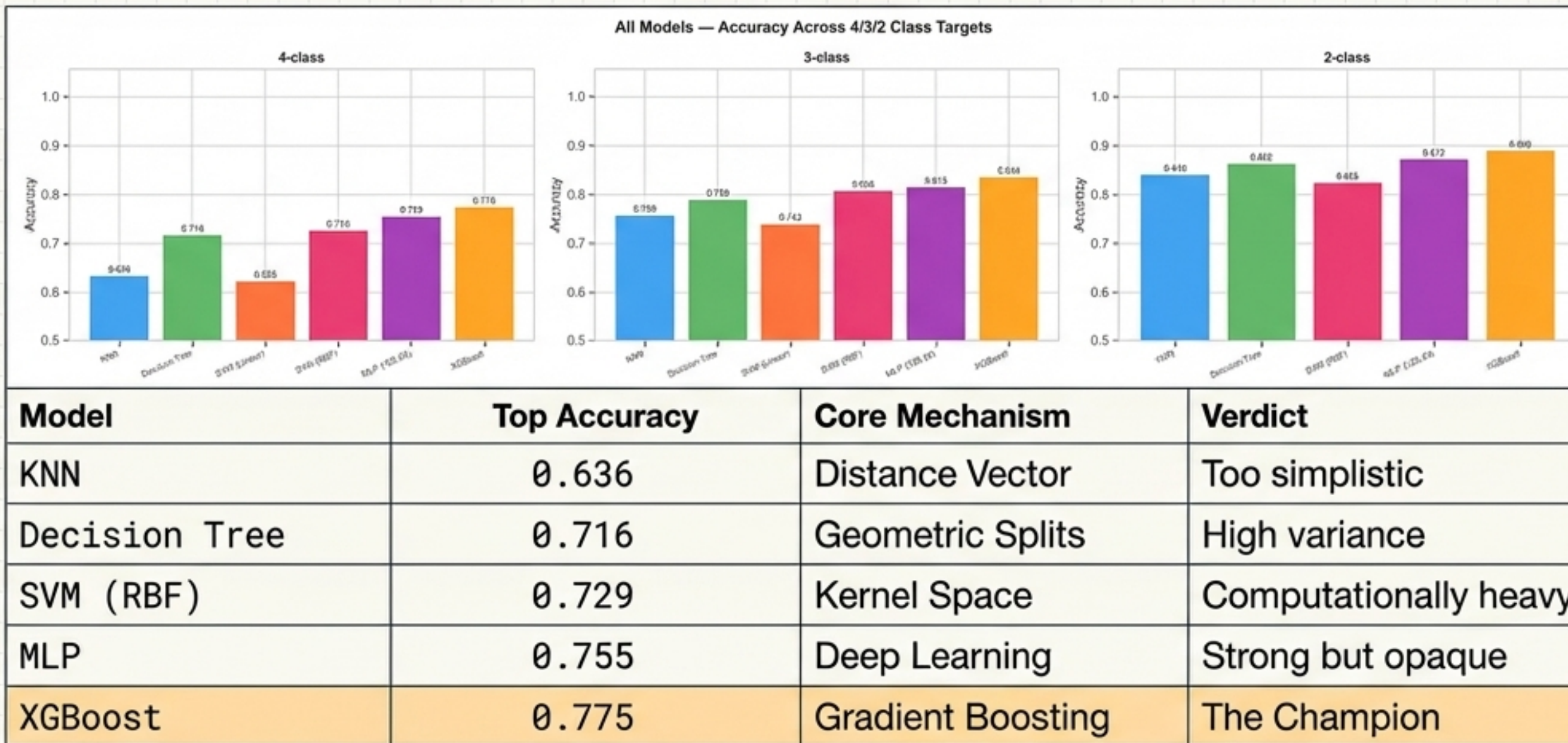
genderXgrip

$$\text{genderEncoded} \times \text{gripForce}$$

Allows the tree algorithms to independently learn distinct, non-linear grip thresholds for men and women.

Insight: Feature engineering adds a critical ~1.8% accuracy boost by mathematically pre-digesting complex physiological relationships for the models.

The ML Bake-off Matrix



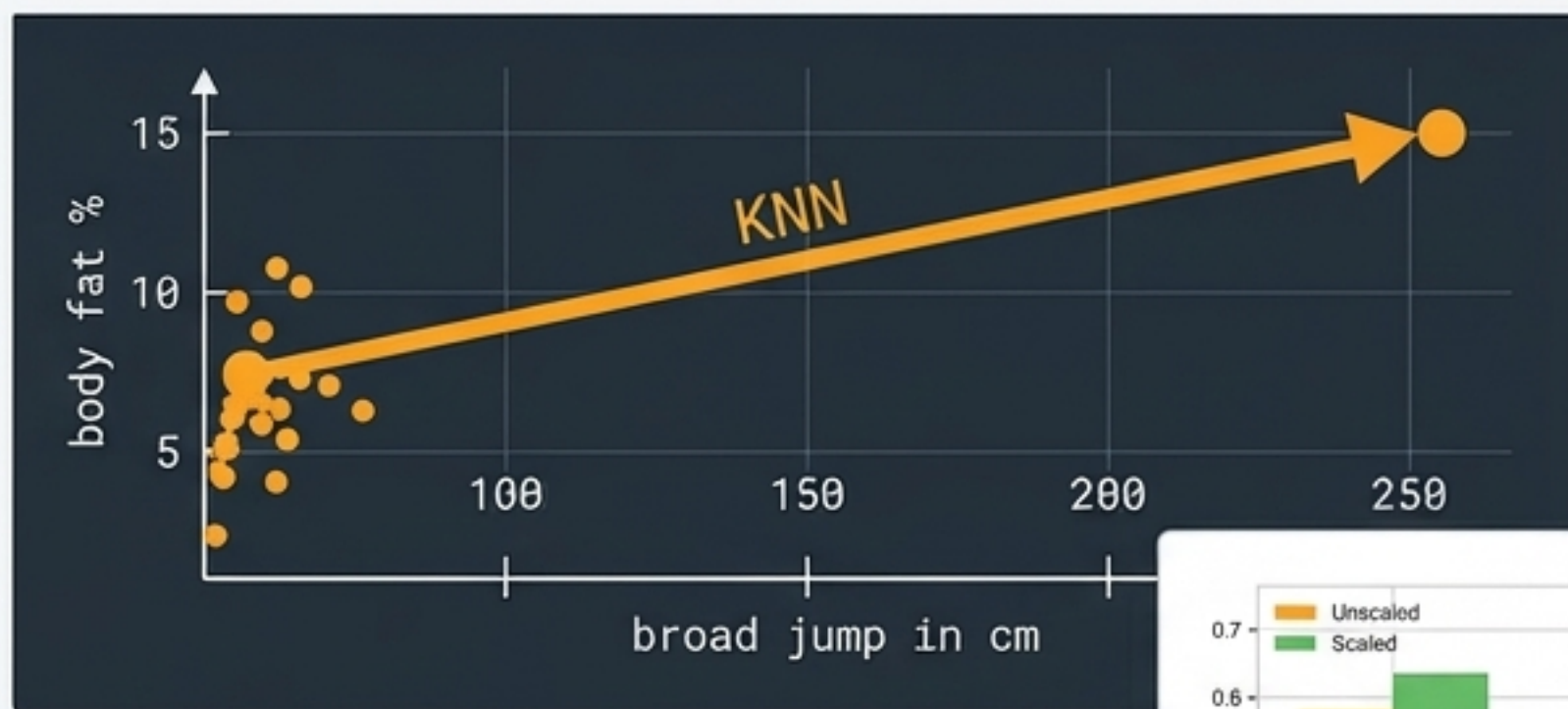
Ensemble Note: A soft voting ensemble (KNN+DT+SVM+MLP) achieved 0.758, proving that combining weak models cannot beat a single, finely tuned XGBoost instance.

Insight: Gradient boosting systematically out-predicts all competitors by leveraging sequential error correction and sophisticated regularization on tabular data.

The Mechanics of Scale

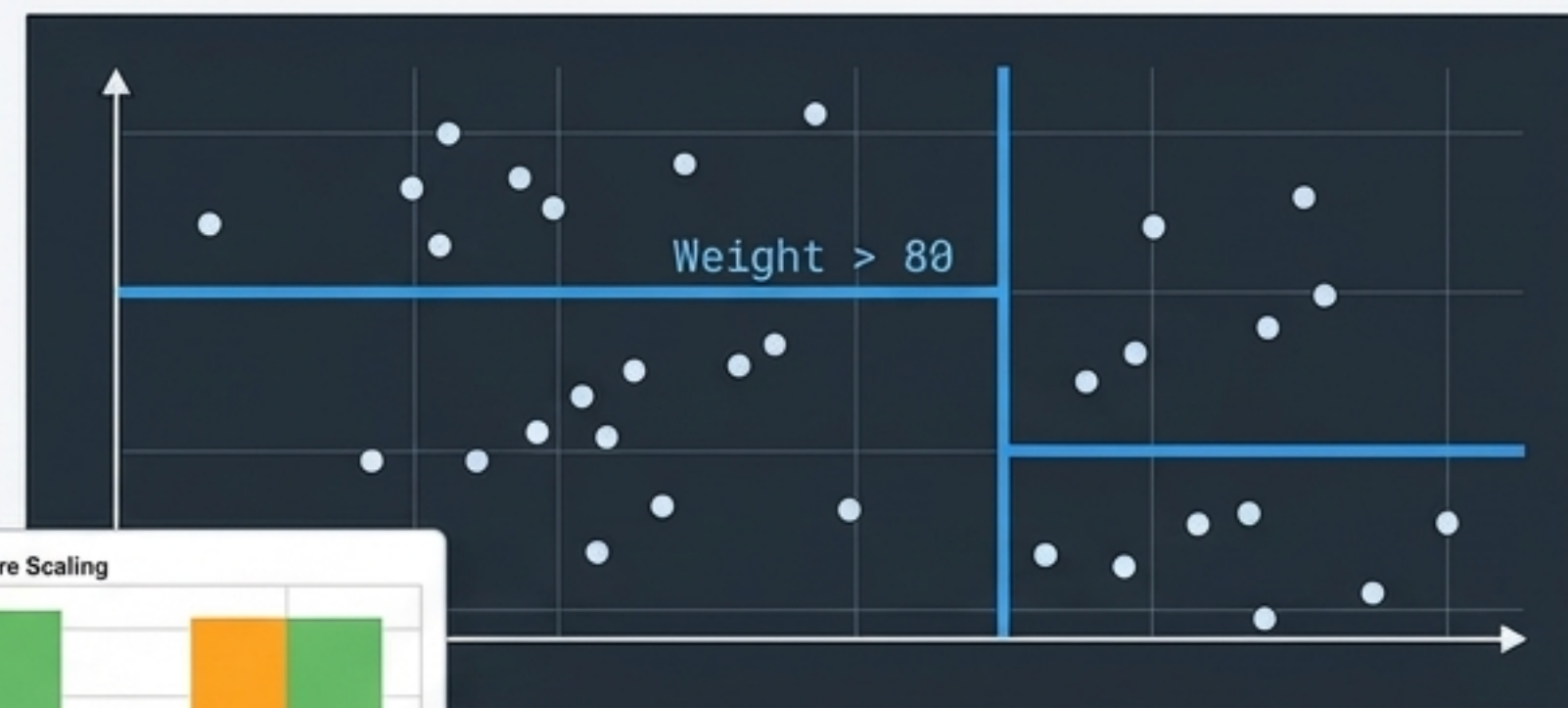
Conceptualizing the Impact of Feature Scaling on Machine Learning Models

Distance Sensitivity

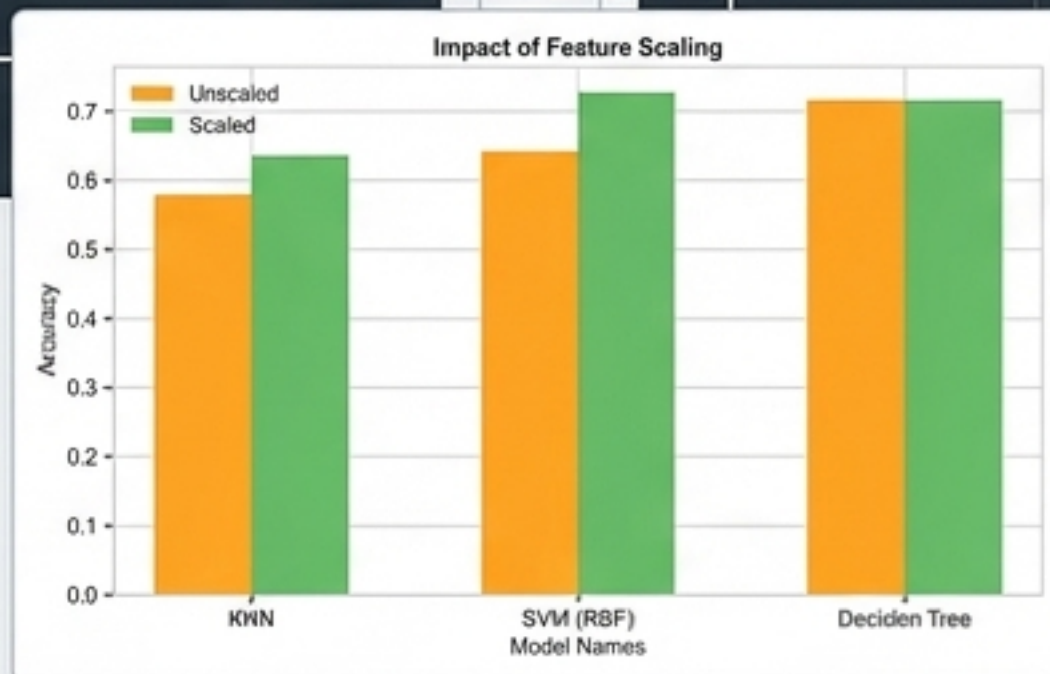


KNN jumped 10-15% in accuracy with feature scaling because distance algorithms are blinded by raw magnitude, causing features with larger ranges to dominate the distance calculation.

Scale Invariance

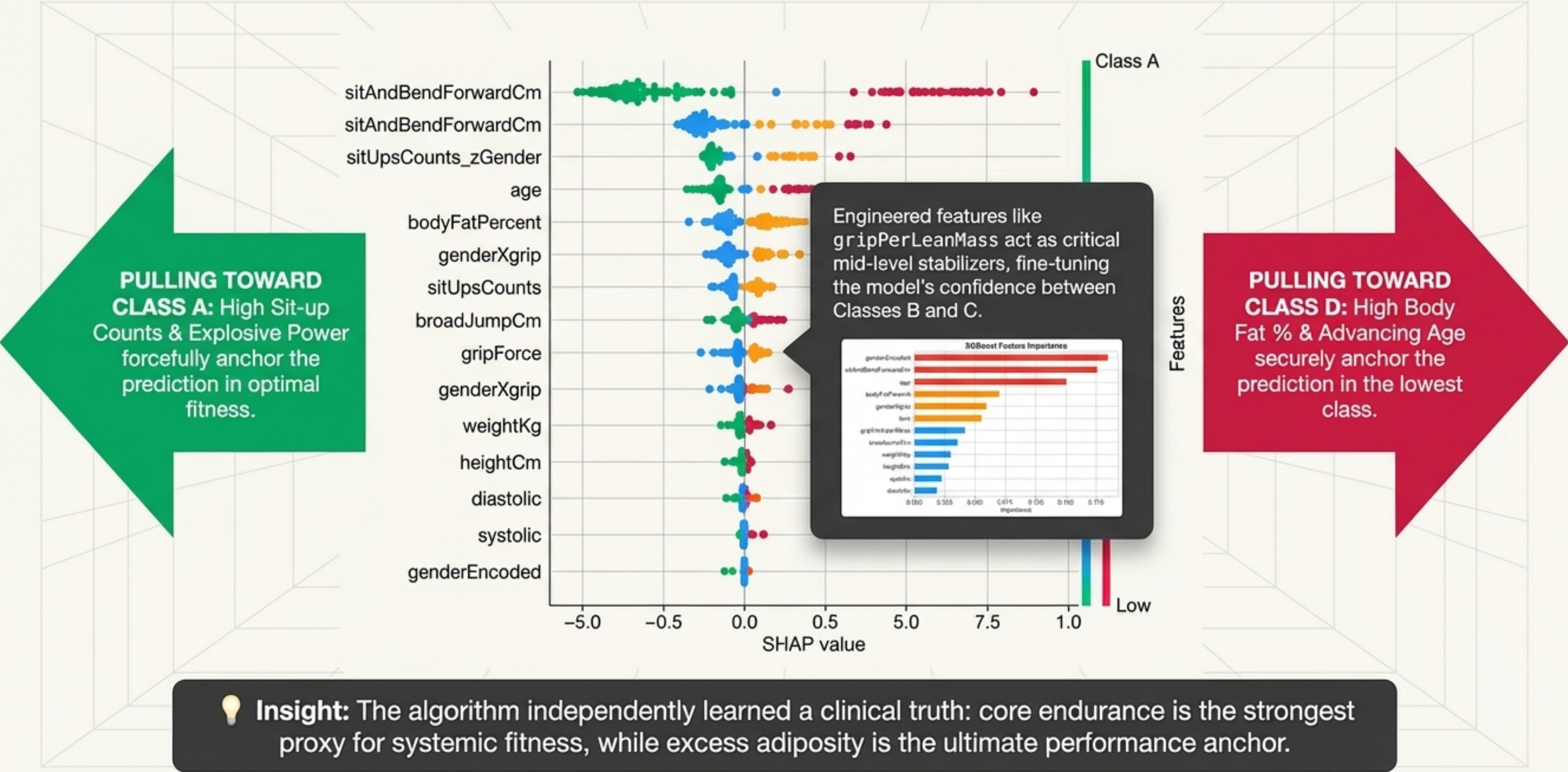


Decision Trees saw exactly 0% change with scaling. Tree splits rely purely on relative ordering (e.g., $x > \text{threshold}$), ignoring raw mathematical scale entirely and focusing on rank.



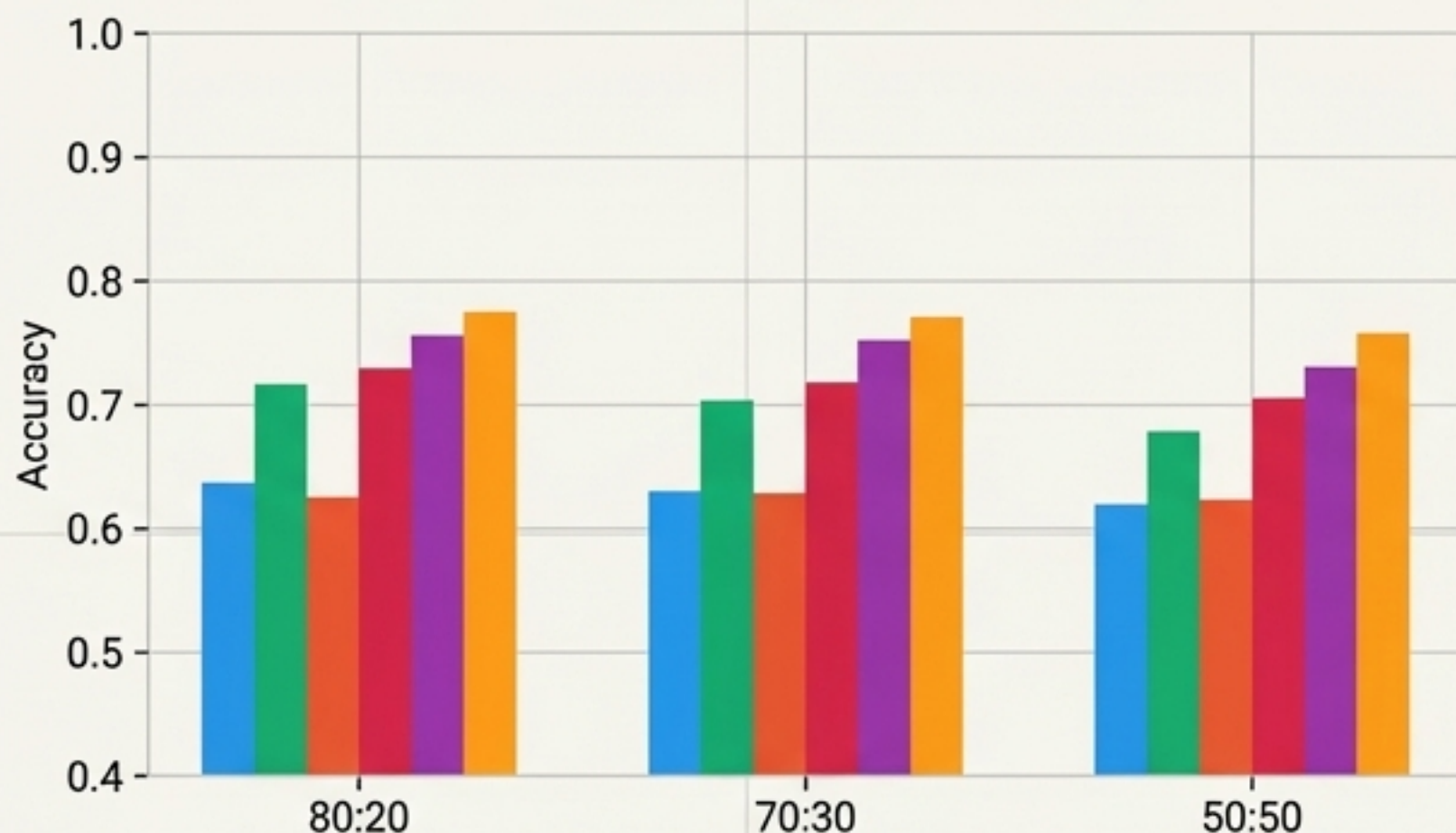
Insight: StandardScaler is a vital prerequisite for spatial models (KNN/SVM/MLP) to achieve peak performance, but **entirely irrelevant and unnecessary** for tree-based architectures which operate on relative ordering.

Anatomy of a Prediction: The Algorithmic Tug-of-War



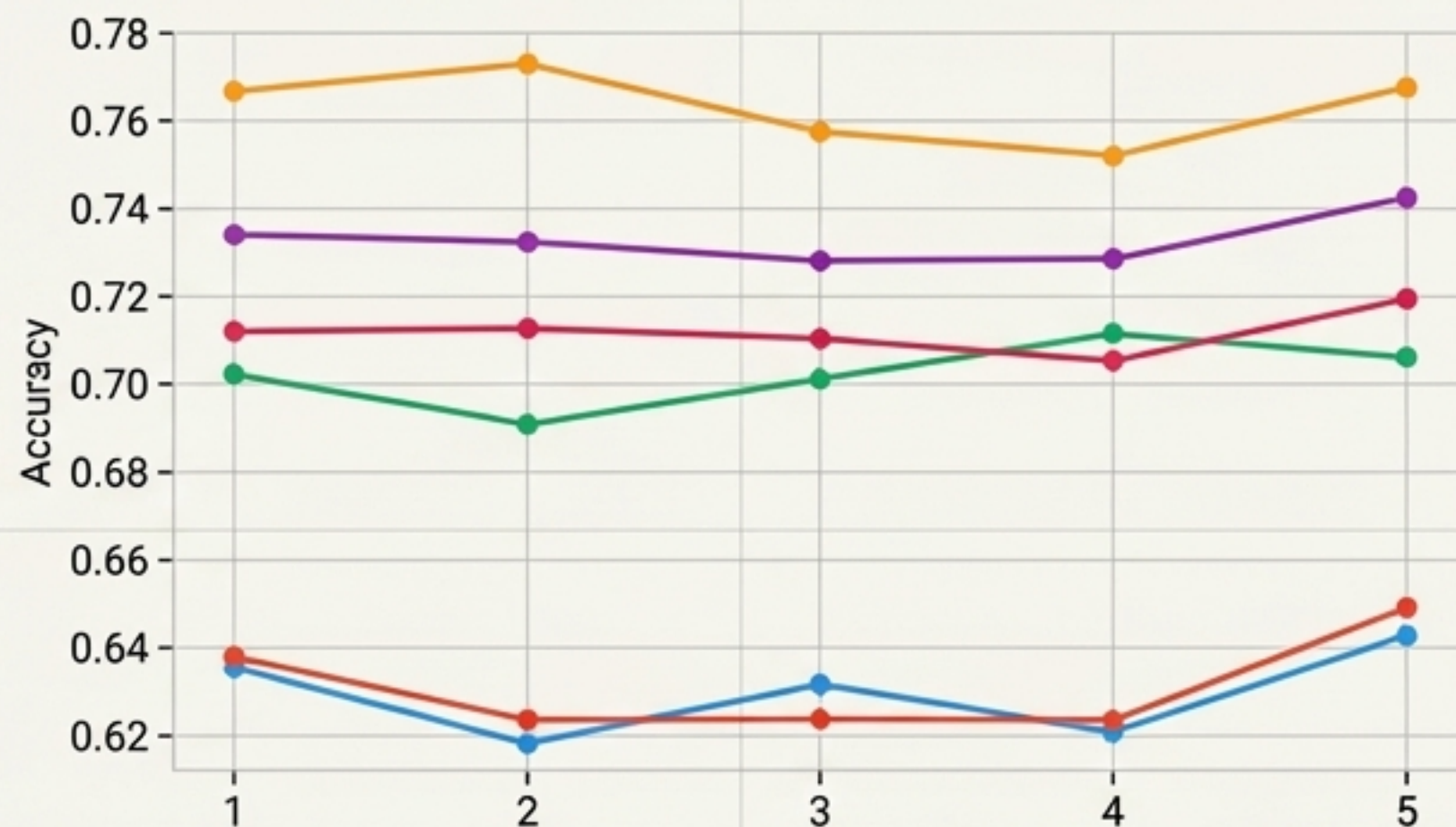
Structural Robustness: Performance Under Pressure

Multi-Split Degradation



XGBoost exhibits iron-clad stability, dropping only -1.1% when aggressively starved of training data (down to 50:50). In contrast, the Decision Tree shows fragile variance, dropping -2.5% under the same conditions.

5-Fold Cross Validation



XGBoost achieves the highest mean CV accuracy (0.763) with tightly clustered variance bands. Paired t-tests confirm this outperformance is statistically significant ($p < 0.05$), proving it is not due to random variation.



Insight: XGBoost does not merely memorize the data; it learns the underlying physiological rules, maintaining strict predictive accuracy even when training data is aggressively constrained.

Synthesis & Conclusions

1.

The Flaw is in the Design

The artificial boundary between Classes B and C is the ultimate bottleneck. Precision fitness tracking requires moving beyond rigid 4-class quartile binning toward overlapping probabilistic models or binary (2-class) pass/fail standards.

2.

Context is King

Raw numbers lie. Feature engineering that mathematically normalizes for gender (`sitUpsCounts_zGender`) and body composition (`gripPerLeanMass`) is essential for unlocking the true predictive signal hidden in biological data.

3.

Gradient Boosting Reigns

For tabular physiological data, XGBoost is unmatched. It natively handles varying scales, extracts signal from overlapping classes, and maintains near-perfect stability under extreme data constraints.

Machine learning cannot fix flawed biological grading systems, but it can expose their fault lines—and predict human potential with unprecedented accuracy when the target is true.